6. [Maximum mark: 6]

The volume of a spherical bubble increases at a constant rate of $5\,\mathrm{cm}^3\,\mathrm{s}^{-1}$.

The initial volume of the bubble can be assumed to be zero.

Find the rate, in $\mathrm{cm\,s}^{-1}$, at which the radius of the bubble is increasing when the volume of the bubble is $20\,\mathrm{cm}^3$.